



Lab 11

PH142 Spring 2026

Midterm Points Recovery Exam

Date: Friday, April 24th

Time: 8:10 – 8:35am (25 minutes)

Location: Dwinelle 155

What to Bring: double-sided cheat sheet, calculator, student ID, pencil/pen

You may earn up to ~10–15 additional points towards your midterm scores.
See Ed post #389 for additional information.



Announcements

- **Lab11:** due 4/24 at 11:59pm
- **Quiz10:** due 4/25 at 12:00pm
- **EC Participation (see Ed #266):** due 4/26 at 10:00pm

Coming up:

- Data project part 3: due 5/1 at 5:00pm



Week 14 Lecture Review

Chi-Square Tests

Goal: To determine whether your data are significantly different from what you expected.

Chi-Square tests are used for categorical data.

There are two types:

- Chi-Square Goodness of Fit
- Chi-Square Test for Independence

Week 14 Lecture Review

Chi-Square Tests

Chi-Square Test Statistic:
$$\chi^2 = \sum_{i=1}^k \frac{(O_i - E_i)^2}{E_i}$$

k = the number of cells in the table.

O = the observed count for the i th group

E = the expected count for the i th group

Week 14 Lecture Review

Chi-Square Goodness of Fit

Goal: To test how well the observed counts “fit” the expected counts.

H_0 : Proportion in observed is the same as the proportion from a given population

H_a : At least one of p_k is not equal to the proportion stated in the null hypothesis

$$p_1 = \#_1, p_2 = \#_2, \dots, p_k = \#_k$$

where $\#_1, \#_2, \dots, \#_k$ are provided or can be derived from percentages provided

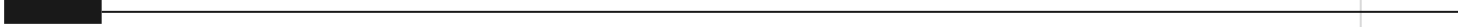
Week 14 Lecture Review

Chi-Square Test for Independence

Goal: To test whether two categorical variables are related to each other.

H_0 : Response and explanatory variables are independent

H_a : Response and explanatory variables are dependent



LAB 11 Walkthrough

Lab Submission

- Follow the directions on the LAB11 file
- Submit using the **Terminal Tab** (next to the console in the bottom left pane)
 - Copy and paste the given line into the terminal
 - Follow prompts (NOTE: the terminal will **not** show your password being typed out!)
- **CHECK IN GRADESCOPE THAT ALL YOUR TESTS PASSED**